

EEG Analysis of Children with Attention-Deficit/Hyperactivity Disorder and Comorbid Reading Disabilities

Adam R. Clarke, Robert J. Barry, Rory McCarthy, and Mark Selikowitz

Abstract

This study investigated electroencephalographic differences between two groups of children with attention-deficit/hyperactivity disorder, combined type, with reading disabilities (ADHD + RD) or without (ADHD) and typical control participants. Twenty participants were included in each group. All participants were between the ages of 8 and 12 years, and groups were matched on age and gender. The electroencephalographic (EEG) was recorded during an eyes-closed resting condition from 21 monopolar derivations, which were clustered into nine regions for analysis. The EEGs were Fourier-transformed to provide absolute and relative power estimates for the delta, theta, alpha, and beta bands. Ratio coefficients were also calculated for the theta/alpha and theta/beta ratios. Compared with controls, the clinical groups demonstrated the increased slow-wave and reduced fast-wave activity commonly reported in the ADHD literature. The ADHD + RD group had more relative theta, less relative alpha, and a higher theta/alpha ratio than the ADHD group. A number of hemispheric differences were also found in the delta and alpha bands. These results suggested that some of the EEG divergences found in the ADHD + RD group represent an electrophysiological component associated with the reading disability that is independent of the EEG divergences found in ADHD.

Attention-deficit/hyperactivity disorder (ADHD) is one of the disorders most commonly treated by child and adolescent psychiatrists in America, with this group of children constituting as much as 50% of child psychiatry clinic populations (Cantwell, 1996). ADHD is a persistent problem that may change with development from preschool through adulthood and interferes with many areas of typical development and functioning in a child's life. However, there is some debate about the exact nature of this disorder. The tenth revision of the *International Classification of Diseases* (ICD-10; World Health Organization, 1993), and the fourth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-IV; American Psychiatric Association, 1994) have listed similar criteria for a disorder characterized by inattention, impulsivity, and

hyperactivity. Whereas the *DSM-IV* recognized the existence of subtypes, allowing a diagnosis based on symptoms either from a combined hyperactive-impulsive dimension or from inattentive behaviors, the ICD-10 required symptoms to be present from each of the three categories of behaviors. These differences have had a significant impact on the reported prevalence of ADHD and on the selection of which children should undergo treatment.

Studies based on *DSM-IV* criteria have found differences in the prevalence of children with ADHD from 4% to 6.1% (Lindgren et al., 1990; Pelham, Gnagy, Greenslade, & Milich, 1992). The *DSM-IV* estimated the prevalence of ADHD in the general population at approximately 3% to 5% of school-age children (American Psychiatric Association, 1994). In Australia, the prevalence rate of ADHD has been estimated

at approximately 2.4% (Gomez, Harrey, Quick, Scharer, & Harris, 1999).

Electrophysiological studies of children with ADHD have typically found an increase in slow-wave activity, primarily in the theta band, compared to controls (Capute, Niedermeyer, & Richardson, 1968; Satterfield, Lesser, Saul, & Cantwell, 1973; Wikler, Dixon, & Parker, 1970). Matousek, Rasmussen, and Gilberg (1984) found that relative delta in posterior regions correlated highly with a diagnosis of minimal brain dysfunction. Mann, Lubar, Zimmerman, Miller, and Muenchen (1992) found that children with ADHD during a resting condition had an increase in absolute amplitude in the theta band that occurred primarily in frontal regions. During cognitive tasks, children with ADHD showed a greater increase in theta activity in frontal and central regions and a de-

crease in beta activity in posterior and temporal regions. Lazzaro et al. (1998) found increased absolute theta in frontal regions and reduced relative beta in the posterior regions during an eyes-open condition.

All of these studies investigated only one subtype of ADHD, although the different editions of the *Diagnostic and Statistical Manual of Mental Disorders* (American Psychiatric Association, 1980, 1987, 1994) all listed multiple subtypes of the disorder. Chabot and Serfontein (1996) studied electroencephalographic (EEG) differences in children with attention-deficit disorder (ADD) with or without hyperactivity using *DSM-III* criteria. The children with ADD were found to have an increase in absolute and relative theta, with the greatest increase being found in frontal regions and at the midline. A slight elevation in relative alpha was also noted, and in a small group of children an increase in beta was also found. Differences between the two subtypes of ADD were found, with the difference appearing to be one of degree of divergence rather than a qualitative difference between the two subtypes.

Kuperman, Johnson, Arndt, Lindgren, and Wolraich (1996), using *DSM-III-R* criteria, studied EEG differences between children with ADHD, children with undifferentiated attention-deficit disorder (UADD), and children without ADD during an eyes-open condition. In relative power, main effects of band were found, with the control group having more delta than the UADD group and less beta than both groups of children with ADD.

Clarke, Barry, McCarthy, and Selikowitz (1998) found EEG differences between children with ADHD, combined type (ADHD-C); ADHD, predominantly inattentive type (ADHD-I); and control children during an eyes-closed resting condition. These differences occurred across all sites in absolute and relative theta and for relative alpha and beta. In posterior regions, relative delta estimates were able to differentiate between all three experimental groups. From these results, it was con-

cluded that the differences between the two subtypes of ADHD reflect a quantitative difference only rather than that the two subtypes are independent in nature.

Although the results of these studies are largely consistent, two opposing interpretations of the data have been proposed. The first interpretation is that the divergence in (EEGs) in children with ADHD represents a *maturational lag* of the central nervous system (Clarke et al., 1998; Mann et al., 1992). The EEG profiles obtained are similar to those of younger children without ADHD, and there is an underlying assumption that they may mature with age. This model has also been supported by studies of the auditory evoked potential (Satterfield, Cantwell, Saul, Lesser, & Podsin, 1973) and of the dominant mean frequency of the EEG (Matsuura et al., 1993) and by behavioral studies (Kinsbourne, 1973).

One of the major problems with this model is that it is not able to explain the existence of ADHD in adults. The maturational lag model implies that as a child with ADHD becomes older, the symptoms of ADHD should lessen. Studies of ADHD in adults have estimated that between 30% and 70% of children with ADHD continue to show symptoms of the disorder as adults (Bellak & Black, 1992) and that although the excessive gross motor activity becomes less, the inattention tends to remain (American Psychiatric Association, 1994).

In contrast to the maturational lag model, a second model holds that ADHD represents a *developmental deviation* of the central nervous system. Chabot and Serfontein (1996) concluded that their results represented a deviation from typical development. This study used a paradigm described by John, Pritchep, Fridman, and Easton (1988) whereby EEG measures were converted to z scores and compared to a normative database. If the measures obtained from a child fell within a statistically determined band derived from a group of younger children without ADHD, the EEG was deemed to

represent a maturational lag. If the results lay outside these parameters, the EEG was viewed as indicating a developmental deviation (John et al., 1983).

Clarke, Barry, McCarthy, and Selikowitz (2001b) compared children with two subtypes of ADHD and a control group, using ratio coefficients between frequency bands and measures of the mean frequency within frequency bands. Results for the theta/alpha and theta/beta ratios showed larger coefficients in the two clinical groups. In the alpha and beta bands, the ADHD groups had a lower mean frequency than controls. In the comparison of the lower frequency bands, the control group had lower mean frequencies in the delta band and higher ratio coefficients for the delta/theta ratio. These results suggested that in the ADHD groups there is a shift in the EEG at both ends of the spectrum toward the theta frequency range, which does not support the maturational lag model.

One problem with most of the studies that have proposed a model of ADHD based on EEG measures is that they have used measures from children with ADHD at one point in time and then postulated what their results meant based on what has been found in the developmental literature for children without ADHD. Bresnahan, Anderson, and Barry (1999) investigated changes in the EEG of individuals with ADHD who were classified into three age groups: children, adolescents, and adults. The results indicated that theta activity remained elevated in adults with ADHD, but that there was a decrease in beta activity with age. From these results, it was concluded that beta activity may be linked to hyperactivity and increased theta activity to impulsivity, although this needs further investigation.

Clarke, Barry, McCarthy, and Selikowitz (2001a) found that EEG power changed at a greater rate in the ADHD-C group than in the ADHD-I group, with power levels between the two groups becoming similar with age. The differences between the ADHD-I

group and the control group remained at a similar level with increasing age. These results suggested that ADHD has two distinct components, the first a hyperactive-impulsive component that matures with age (i.e., a maturational lag) and the second an inattentive component that remains stable with age (i.e., a developmental deviation).

All of these studies have investigated the EEGs of individuals with ADHD without investigating the possible impact of other comorbid factors. For example, the previous ADHD studies from our laboratory have excluded individuals with any signs of comorbid psychiatric disorders. The presence of reading disabilities (RD) in children with ADHD has been estimated at 20% to 25% (Pliszka, 1998), although values in the literature have ranged from 10% to 92% (Biederman, Newcorn, & Sprich, 1995). Despite the high levels of comorbidity found for ADHD and RD, a number of researchers have argued that the two disorders are distinct (Pliszka, 1998; Shaywitz & Shaywitz, 1991).

EEG studies of children with RD have found that 87% of such children had EEG divergences (John et al., 1977). Lubar et al. (1985) found that children with RD had excessive levels of theta activity in the 4- to 8-Hz range. In the parieto-occipital regions, excessive delta and theta wave activity have been found, along with deficiencies in relative alpha (Ahn et al., 1980; Chabot, Merkin, Wood, Davenport, & Serfontein, 1996). A number of studies have further found increased slow-wave activity in the left hemisphere, particularly in the posterior regions. Rebert, Wexler, and Sproul (1978) found greater theta power in the left parietal region compared to the right parietal region during an eyes-open resting condition. Duffy, Denckla, Bartels, and Sandini (1979) found differences between individuals with and without dyslexia, primarily in the left posterior regions. Harmony et al. (1990) found increased absolute delta and theta in RD groups, primarily in the left hemisphere. Delta wave activity is dominant for the first few months of life, but it is quickly re-

placed with theta and alpha activity as control shifts from subcortical to cortical structures (Porges & Fox, 1986), although random posterior delta and theta activity are common during childhood and adolescence (Fisch, 1991). Both delta and theta activity have been associated with underarousal, and focal delta activity can indicate a structural divergence (Andreassi, 1995). These results indicate atypical functioning in the left posterior region of the brain.

Such results are supportive of those results from studies that used other imaging techniques to investigate brain functioning in children with RD. Positron emission tomography (PET) studies involving language tasks have found differences in the left hemisphere of individuals with RD compared to individuals without RD (Flowers, 1993). Cerebral blood flow (CBF) studies (Flowers, 1993) have shown that there is an inferior left parietal component associated with word meaning in children with RD. Galaburda, Sherman, Rosen, Aboitiz, and Geschwind (1985) and Kaufman and Galaburda (1989) also found that the brains of individuals with RD had significantly more focal dysplasias, particularly in the language regions that border the sylvian fissure, compared with those of controls. In a review of neuroimaging studies, Pugh et al. (2000) suggested that fluent reading is related to the correct functioning of a dorsal circuit and a ventral circuit in the posterior region of the left hemisphere. This posterior system is functionally disrupted in developmental dyslexia. Relative to average readers, individuals with RD demonstrate heightened reliance on both inferior frontal and right hemisphere posterior regions, presumably in compensation for the left hemisphere posterior difficulties.

Few studies have investigated EEG differences in ADHD children with and without comorbid RD. Ackerman, Dykman, Oglesby, and Newton (1994) investigated EEG differences between children with ADHD, ADHD with learning delays, and ADHD with dys-

lexia, using seven cognitive tasks. The ADHD group had greater parietal and midline low-frequency beta than the other two groups. In a follow-up study, Ackerman, Dykman, Oglesby, and Newton (1995) found that both dysphonetic and phonetic poor readers (who may also have had a diagnosis of ADHD) had less parietal beta power than an ADHD control group during reading tasks. These results were seen as evidence of less activation in the posterior regions. This study did not investigate resting state EEG differences between groups. Ackerman et al. also used mixed groups of children with the hyperactive and inattentive subtypes of ADHD, which have been found to have different EEG profiles (Chabot & Serfontein, 1996; Clarke et al., 1998, 2001a, 2001b). The aim of the present study was to investigate whether EEG differences can be found between children with ADHD with and without comorbid RD and to quantify the nature of such differences during a resting condition.

Method

Participants

This study was conducted over a 3-year period. Three groups of 20 children, with 18 boys and 2 girls in each group, participated in this study. All children were between the ages of 8 and 12 years and were right-handed and right-footed. Participants had a full scale IQ score of 85 or higher. The children were classified into a group with ADHD and comorbid RD (ADHD + RD), a group with ADHD only, and a control group. Both clinical groups of children were drawn from new patients presenting for an assessment for ADHD at a pediatric practice in Sydney, Australia. This practice receives referrals from the entire state of New South Wales and covers all socioeconomic groups. The clinical group members had not been diagnosed as having ADHD previously, had no history of using medication for the disorder, and were tested before being prescribed any medication. Thus, this

specialist medical practice was the first treatment point for these children, as is common in Australia. All children between the ages of 8 and 12 years were invited to participate in the study, with no patients refusing. Consecutive patients were used as participants until all cells in the experimental design were filled. Randomization was not feasible in this study due to specifics of age, gender, diagnosis, and exclusion criteria. The control group consisted of children from local schools and community groups.

Inclusion in the clinical groups was based on clinical assessments by a pediatrician and a psychologist; children were included only when both professionals agreed on the diagnosis. *DSM-IV* criteria were used, and children were included if they met the full diagnostic criteria for the diagnosis of attention-deficit/hyperactivity disorder, combined type. A structured clinical interview was used that incorporated information from as many sources as were available. The interview included a description of the presenting problem and a medical history given by a parent or guardian, a physical examination, an assessment for neurological soft signs, a review of school reports for the past 12 months looking for behavioral/learning problems, reports from any other health professionals, and behavioral observations during the assessment. Children were also assessed using the *Wechsler Intelligence Scale for Children-Third Edition* (WISC-III; Wechsler, 1991), the *Neale Analysis of Reading* (Neale, 1988), and the *Wide Range Achievement Test-Revised* (WRAT-R; Jastak & Wilkinson, 1984) spelling test. The *Conners Rating Scale* (Conners, 1990) was also used. Children were excluded from the clinical groups if they had a history of a problematic prenatal, perinatal, or neonatal period, a disorder of consciousness, a head injury with cerebral symptoms, a history of central nervous system diseases, convulsions or a history of convulsive disorders, a paroxysmal headache, or tics.

For inclusion in the ADHD group, participants had to score in the average

range on the measures of accuracy and comprehension on the *Neale Analysis of Reading* and have a standard score of 90 or more on the WRAT-R spelling test. Inclusion in the ADHD + RD group was based on the children scoring 2 years or more below their chronological age for accuracy and comprehension of reading and having a standard score of 75 or less for spelling.

Inclusion of children in the control group was based on an uneventful prenatal, perinatal, and neonatal period and the absence of disorders of consciousness, head injury with cerebral symptoms, a history of central nervous system diseases, obvious somatic diseases, convulsions or a history of convulsive disorders, paroxysmal headache, enuresis or encopresis after the fourth birthday, tics, stuttering, pavor nocturnes, excessive nailbiting, obvious mental diseases, conduct disorders, or any deviation with regard to mental and physical development. Control participants also had to score in the average range on the measures of accuracy and comprehension on the *Neale Analysis of Reading* and have a standard score of 90 or more on the WRAT-R spelling test. Assessment for inclusion as a control participant was based on a clinical interview with a parent or guardian that was similar to that of the ADHD participants, using the same sources of information and the same psychometric assessments used for the clinical group participants.

Any children who showed signs of depression, anxiety, oppositional behavior, or syndromal disorders were excluded from this study. Children were also excluded from all groups if spike wave activity was present in the EEG. Across all groups, participants were matched on gender and on age within 1 year.

Procedure

The ADHD and control groups were tested in a single session lasting approximately 2.5 hours. Participants were first assessed by a pediatrician, who performed a physical examination and took a clinical history. Participants then

had a psychometric assessment consisting of a WISC-III, *Neale Analysis of Reading*, and WRAT-R spelling. At the end of this assessment, participants had an electrophysiological assessment consisting of evoked potentials and an EEG. The EEG was recorded at the end of the session in an eyes-closed resting condition, while participants were seated in a reclining chair. Electrode placement was in accordance with the international 10-20 system, using an ElectrocapTM produced by Electrocap International. The activity in 21 derivations was divided into nine regions by averaging in each region. These regions were the left frontal (Fp1, F3, F7), midline frontal (Fpz, Fz), right frontal (Fp2, F4, F8), left central (T3, C3), midline central (Cz), right central (T4, C4), left posterior (T5, P3, O1), midline posterior (Pz, Oz), and right posterior (T6, P4, O2). A single electro-oculogram (EOG) electrode, referenced to Fpz, was placed beside the right eye, and a ground lead was placed on the left cheek. A linked ear reference was used with all EEG, and reference and ground leads were 9-mm tin disk electrodes. Impedance levels were set at less than 5 k Ω .

The EEG was recorded and Fourier-transformed by a Cadwell Spectrum 32TM, software version 4.22, using test type EEG, montage Q-EEG. The sensitivity was set at 150 microvolts per centimeter, low-frequency filter 0.53 Hz, high-frequency filter 70 Hz, and 50 Hz notch filter. The sampling rate of the EEG was 200 Hz and the Fourier transformation used 2.5-second epochs.

Thirty 2.5-second epochs were selected from the live trace and stored to floppy disk. Epoch rejection was based on both visual and computer selection. Computer rejection levels were set using a template recorded at the beginning of the session, and all subsequent epochs were compared to this. The EOG rejection level was set at 50 microvolts. The technician also visually appraised every epoch and decided to accept or reject it. The stored data were further reduced by a second technician to 24 epochs (1 minute) for Fourier analysis. The EEG was ana-

lyzed in four frequency bands: delta (1.5–3.5 Hz), theta (3.5–7.5 Hz), alpha (7.5–12.5 Hz), and beta (12.5–25 Hz), for both absolute and relative power, as well as the total power of the EEG (1.5–25 Hz). The mean frequency was calculated for each band and for the total EEG. Ratio coefficients were also calculated between frequency bands by dividing the power of the lower frequency band by the power of the higher frequency band. These were calculated for theta/alpha and theta/beta frequencies.

Statistical Analysis

ANOVA was performed examining the effects of region and group for each band in absolute and relative power, total power, and ratio coefficients. The effects of region were examined in two orthogonal three-level repeated-measures factors. The first of these was a sagittal factor, within which planned contrasts compared the frontal regions with the posterior regions and the mean of the frontal and posterior regions with the central regions. The second factor was laterality, within which planned contrasts compared activity in the left hemisphere with that in the right hemisphere and the mean of the two hemispheres with the midline region. These planned contrasts allowed optimal clarification of site effects within the regions studied.

Within the group factor, planned contrasts compared the clinical groups with the control group (to establish ADHD differences from controls) and the ADHD + RD group with the ADHD group. As all these contrasts were

planned and there were no more of them than the degrees of freedom for effect, no Bonferroni type adjustment to α was required (Tabachnick & Fidell, 1989). We point out that analyses are carried out over a substantial number of variables, each of which may be considered to constitute a separate experiment. It should be noted that this increases the frequency of Type I errors. As this is an increase in frequency rather than in probability, it cannot be controlled by adjustment of α levels (Howell, 1997). For space reasons, only between-group effects and interactions are reported here.

Results

As expected from the group inclusion criteria, significant differences were found between the two clinical groups (see Table 1) on the measures of reading accuracy, $F(1, 38) = 73.33, p < .001$, reading comprehension, $F(1, 38) = 39.27, p < .001$, and spelling, $F(1, 38) = 149.55, p < .001$.

EEG group differences are shown in Figures 1 (sagittal view) and 2 (lateral view). In absolute delta (see Figures 1 and 2), a significant hemispheric difference in topography was found between the two ADHD groups, with the maximal difference between groups occurring in the left posterior regions, $F(1, 57) = 5.94, p < .05$. For relative delta, the two ADHD groups had more activity than the control group, $F(1, 57) = 5.65, p < .05$. This was more pronounced in the posterior regions compared to the frontal regions, $F(1, 57) = 7.28, p < .001$. The ADHD + RD group had more power than the ADHD

group in the posterior regions compared to the frontal regions, $F(1, 57) = 5.41, p < .05$. The difference in power between the two ADHD groups and the control group was greater in the left hemisphere than in the right hemisphere, $F(1, 57) = 4.2, p < .05$, and the difference between the ADHD + RD and ADHD groups was greater in the right hemisphere than in the left, $F(1, 57) = 8.57, p < .01$.

In absolute theta (see Figures 1 and 2), the two ADHD groups had significantly more power than the control group, $F(1, 57) = 3.99, p < .05$. Significant differences were also found in the relative theta band, with the ADHD groups having more theta than the control group, $F(1, 57) = 47.17, p < .001$, and the ADHD + RD group having more theta than the ADHD group, $F(1, 57) = 7.16, p < .01$. The difference between the midline and the two hemispheres was greater in the two ADHD groups than in the control group, with the maximal group difference occurring in the frontal regions, $F(1, 57) = 4.61, p < .05$.

In absolute alpha (see Figure 1), the two ADHD groups had less power than the control group, $F(1, 57) = 7.27, p < .01$, and this group difference was greater in posterior regions than in frontal regions, $F(1, 57) = 7.27, p < .01$. The difference between central regions and the mean of the frontal and posterior regions was also smaller in the clinical groups, $F(1, 57) = 6.72, p < .05$, which indicated that they had a lesser posterior enhancement in power than the control group. The difference in power between the midline and the two hemispheres (see Figure 2) was smaller in the two clinical groups than in the control group, $F(1, 57) = 6.65, p < .05$. In comparison to the ADHD group, the ADHD + RD group had a smaller enhancement of power in the right hemisphere compared to the left hemisphere, $F(1, 57) = 5.97, p < .05$, and this group difference was greatest in the posterior regions, $F(1, 57) = 4.98, p < .05$. For relative alpha, the two clinical groups had less activity than the control group, $F(1, 57) = 35.93, p < .001$,

TABLE 1
Group Differences in the Measures of Accuracy, Comprehension, and Spelling

Group	Chronological age ^a	Accuracy ^a	Comprehension ^a	Spelling ^b
Control	10.77	11.55	11.50	107.95
ADHD	10.78	11.41	11.17	104.40
ADHD + RD	10.80	7.55	8.06	71.75

Note. $n = 20$ for all groups.

^aIn years. ^bStandard score.

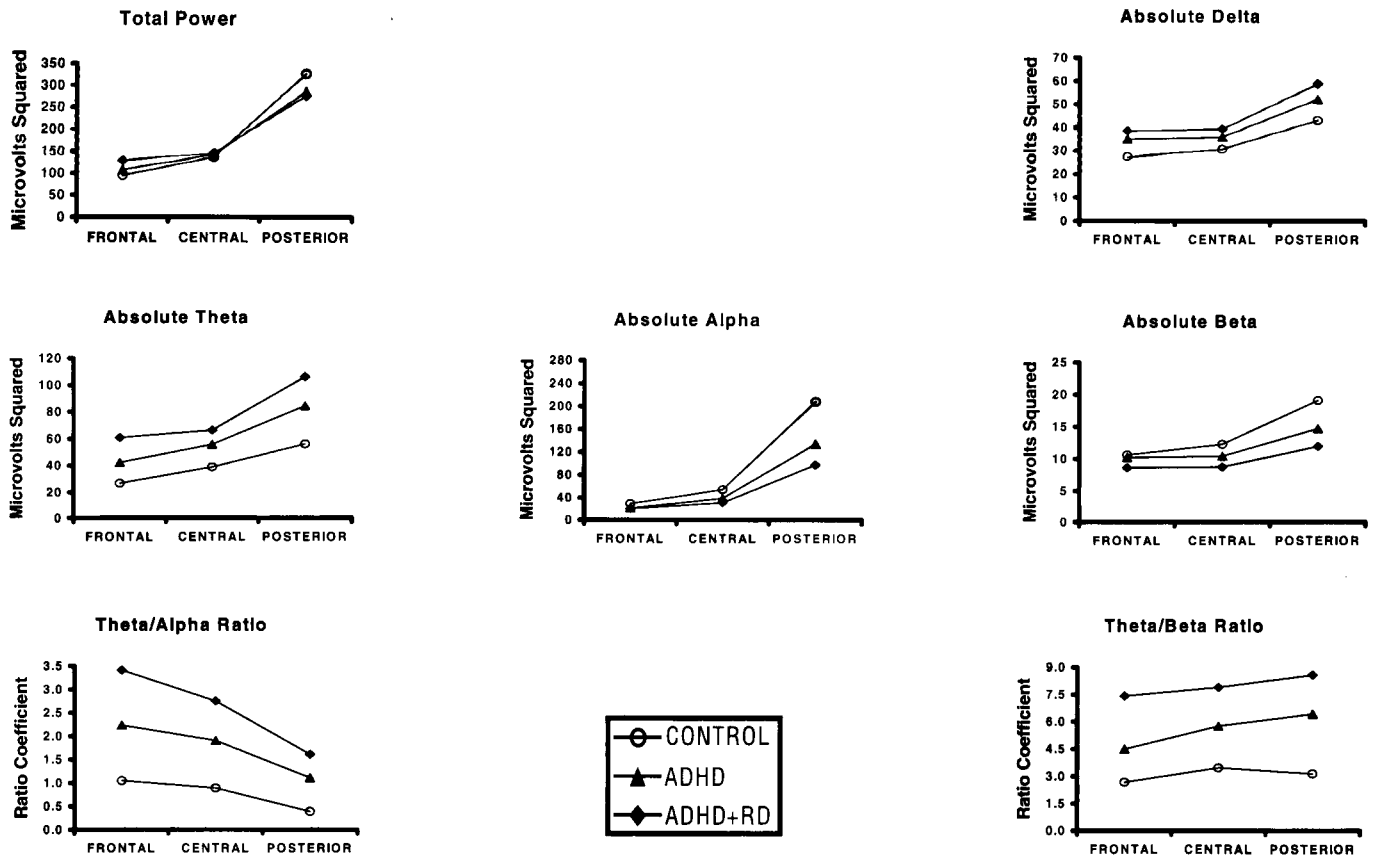


FIGURE 1. Frequency distribution and ratio coefficients as a function of scalp region, sagittal section from frontal to posterior regions, for absolute delta, absolute theta, absolute alpha, absolute beta, theta/alpha ratio, theta/beta ratio, and total power.

and the ADHD + RD group had less alpha than the ADHD group, $F(1, 57) = 4.33$, $p < .05$. The difference in power between the two clinical groups and the control group was greater in the posterior regions than in the frontal regions, $F(1, 57) = 11.61$, $p < .001$. In the clinical groups, the midline had less power than the two hemispheres in the frontal regions and greater power than the two hemispheres in the posterior regions, $F(1, 57) = 4.79$, $p < .05$, whereas in the control group, the midline power was greater than the power in the two hemispheres in all regions. The reduced power found in the ADHD + RD group compared to the ADHD group was greater in the left hemisphere than in the right hemisphere, $F(1, 57) = 4.19$, $p < .05$.

In absolute beta (see Figures 1 and 2), the two clinical groups had less power than the control group, $F(1, 57) = 4.41$,

$p < .05$, with this difference being greater in the posterior regions than in the frontal regions, $F(1, 57) = 5.04$, $p < .05$. For relative beta, the clinical groups had less power than the control group, $F(1, 57) = 5.00$, $p < .05$, and this difference was greater in the frontal regions than in the posterior regions, $F(1, 57) = 9.7$, $p < .01$. A group difference for laterality was found, with relatively greater power occurring in the right hemisphere than in the left hemisphere, $F(1, 117) = 6.34$, $p < .05$, in the two clinical groups compared with the control group, $F(1, 57) = 4.11$, $p < .05$.

Significant differences were found for the theta/alpha ratio (see Figure 1), with the clinical groups having a higher ratio than the control group, $F(1, 57) = 22.16$, $p < .001$, and the ADHD + RD group having a higher ratio than the ADHD group, $F(1, 57) = 6.12$, $p < .05$. Both of these differences

were greater in the frontal regions than in the posterior regions, $F(1, 57) = 9.09$, $p < .01$, and $F(1, 57) = 16.65$, $p < .001$, respectively. The difference in ratios between the midline and the two hemispheres (see Figure 2) was greater in the clinical groups than in the control group, $F(1, 57) = 12.04$, $p < .001$, and this group difference was greater in the frontal regions than in the posterior regions, $F(1, 57) = 6.87$, $p < .05$. The clinical groups had a greater theta/beta ratio (see Figure 1) than the control group, $F(1, 57) = 10.35$, $p < .01$. The difference in ratios between the midline and the two hemispheres (see Figure 2) was greater in the clinical groups than in the control group, $F(1, 57) = 8.18$, $p < .01$, and this difference was greater in the frontal regions than in the posterior regions, $F(1, 57) = 4.41$, $p < .05$. No group differences were found for total power.

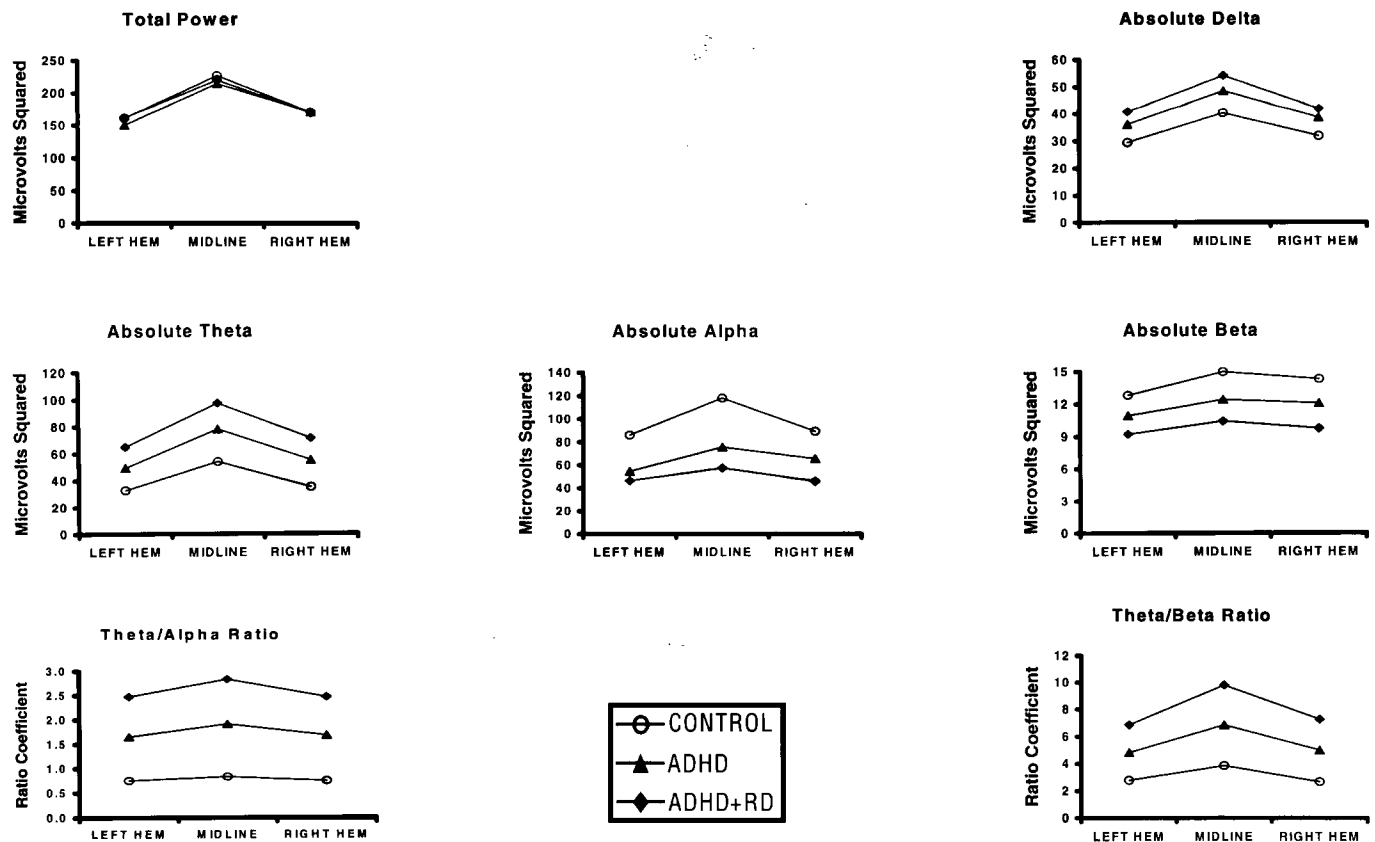


FIGURE 2. Frequency distribution and ratio coefficients as a function of scalp region, lateral section from left to right hemisphere, for absolute delta, absolute theta, absolute alpha, absolute beta, theta/alpha ratio, theta/beta ratio, and total power.

Discussion

Children with ADHD have been found to have elevated levels of slow-wave activity when compared with typical children (Capute et al., 1968; Clarke et al., 1998, 2001; Satterfield & Cantwell, 1974; Satterfield, Cantwell, et al., 1973). This slow-wave activity primarily occurs in the theta band in frontal regions (Bresnahan et al., 1999; Lubar, 1991; Mann et al., 1992; Matousek et al., 1984) and at the midline (Chabot & Serfontein, 1996). Decreased levels of relative alpha and beta have also been found in individuals with ADHD in comparison to control individuals (Clarke et al., 1998, 2001a, 2001b; Mann et al., 1992). Analyses of ratio coefficients between frequency bands, in particular of the theta/alpha and theta/beta ratios, have demonstrated good utility in measuring maturational changes in the EEG of typical children

(Matousek & Petersen, 1973; Matousek et al., 1984) as well as differences between children with and without ADHD (Clarke et al., 2001a, 2001b; Janzen, Graap, Stephanson, Marshall, & Fitzsimmons, 1995; Lubar, 1991).

In the present study, the clinical groups showed less absolute and relative alpha and beta, more absolute theta, and more relative delta and theta than the control group. Regionally, the clinical groups also had less absolute alpha and beta and less relative delta, theta, and alpha in the posterior regions, and less relative beta in the frontal regions, than the control group. The theta/alpha and theta/beta ratios also differentiated between the clinical groups and the control group. These results demonstrate that increased theta activity, decreased alpha and beta, especially in the posterior regions, and increased theta/alpha and theta/beta ratios provide consistent, replicable

differences between children with and without ADHD.

EEG studies of children with RD have found excessive delta and theta wave activity in the parieto-occipital regions, along with deficiencies in relative alpha (Ahn et al., 1980; Chabot et al., 1996). Greater relative theta and increased left hemisphere delta have also been documented (Harmony et al., 1990; Lubar et al., 1985).

The present study found that globally, the ADHD + RD group had more relative theta, less relative alpha, and a higher theta/alpha ratio than the ADHD group. Regionally, the ADHD + RD group had more posterior relative delta and a higher theta/alpha ratio in frontal regions when compared with the ADHD group. Note that these differences were in EEG frequency bands that were found to differ between the clinical groups and the control group. Some researchers have suggested that

the EEG differences between ADHD subtypes are only quantitative in nature and that children with the highest level of impairment also have the highest level of divergence in their EEGs (Chabot & Serfontein, 1996; Clarke et al., 1998). As the ADHD + RD group have the same inattentive and behavioral problems as the ADHD group, in addition to RD, the higher level of EEG deviance in these frequency bands could be considered an enhancement of the electrophysiological divergences found in the ADHD group children.

Several researchers have suggested that although RD is often found in children with ADHD, the disorders are independent (Pliszka, 1998; Shaywitz & Shaywitz, 1991). Clarke et al. (2001a) investigated maturational changes in the EEG of children with inattentive and combined subtypes of ADHD. The children with the combined subtype had similar but more extreme EEG divergences than the children with the inattentive subtype. However, when age-related changes were measured, the results indicated that hyperactive-impulsive and inattentive components were reflected in the EEG and that the hyperactive-impulsive component overlay an additional level of EEG divergence on the inattentive component, although the two components were independent. From the present results, it appears that the extra EEG divergences found in the ADHD + RD group represent another electrophysiological component that is associated with RD and is probably independent of the ADHD diagnosis. This is supported by the fact that the differences found between the two clinical groups in this study are similar to the EEG divergences found in other studies that compared children with RD and children without disabilities.

Several studies have found EEG divergences in the left posterior region in children with RD (Duffy et al., 1979; Rebert et al., 1978). These differences have primarily involved an increase in absolute delta and theta activity (Harmony et al., 1990). Such results are supportive of results from studies that

used other imaging techniques to investigate brain functioning in children with RD (Hynd & Semrud-Clikeman, 1989). PET studies involving language tasks have found differences in the left hemisphere of children with RD compared to children without RD (Flowers, 1993). CBF studies (Flowers, 1993) have shown that there is an inferior left parietal component associated with word meaning in children with RD. In the present study, the ADHD + RD group also showed higher left posterior absolute delta than the ADHD group. Delta activity has been associated with underarousal (Andreassi, 1995), which is consistent with the studies that have found decreased CBF in the left parietal region (Flowers, 1993). These results further support the findings of atypical functioning in the left posterior region of the brain in RD.

Neuroimaging studies have suggested that the right posterior region may compensate for the left hemispheric posterior difficulties in children with RD (Pugh et al., 2000). The right posterior regions have been found to be associated with affective components of language (Bhatnagar & Andy, 1983) and with narrative passage reading (Hynd, Hynd, Sullivan, & Kingsbury, 1987), indicating right posterior involvement in some aspects of typical reading. The present results are consistent with these findings, as specific divergences were found in the EEG in the right posterior region. The ADHD group showed more right hemisphere relative delta and less relative alpha. The ADHD + RD group showed less absolute alpha in the right hemisphere than the ADHD group, with the difference being maximal in the right posterior region. Relative power is a percentage estimate of how much the power in a particular band contributes to the total power in the EEG, and the increased relative delta is probably partially associated with the decreased alpha. Alpha activity is dominant in the posterior regions during an eyes-closed resting condition while the person is alert. This has been viewed as a possible alert but resting state of the

visual cortex (Fisch, 1991), as alpha activity is blocked when a person becomes drowsy. These results would further suggest underarousal in the right posterior region as well as in the left. This would mean that although the right posterior region may provide some compensatory functioning in children with RD, it is also showing deficits in functioning.

The number of hemispheric differences found between the ADHD + RD and ADHD groups further supports the independence of some components in the two groups. In Clarke et al. (1998, 2001a, 2001b), few hemispheric differences were found between the ADHD groups. However, with the RD group, the number of hemispheric differences increased, especially in the posterior regions. These results support the hypothesis that ADHD and RD have distinct electrophysiological components.

From these results, it appears that when a child has multiple comorbid factors, the EEG results obtained represent an accumulation of these different components, with each component overlaying a further level of EEG divergence. This indicates that EEG measures have potential for use in the diagnosis of some of these disorders. The reliability and specificity of these factors needs to be further examined at an individual level for their clinical use to be properly evaluated.

As an initial investigation, this study clustered children on the basis of a lag between reading ability and age. However, it is possible that this sample could represent a heterogeneous group of children with RD. Other studies have found that there are distinct subtypes of children with RD, based on different types of deficits (Golden & Zenhausen, 1983; Watson, Goldgar, & Ryschon, 1983). Further studies should investigate possible differences in the EEGs of children with these subtypes.

This study found EEG differences between children with ADHD and children with ADHD and RD. Widespread differences were found in relative theta and alpha, a higher theta/

alpha ratio, and posterior differences in relative delta. Hemispheric differences were also found in relative delta and alpha and in absolute delta in the left posterior region. These results suggested that RD has some components that are distinct from those found in children with ADHD, although the two disorders commonly occur together. The resultant EEGs from ADHD + RD children presented as an accumulation of the EEG anomalies associated with each component of the disorders that were present. The specificity of these EEG associates of ADHD and RD may support the diagnostic use of EEG in this field.

ABOUT THE AUTHORS

Adam R. Clarke, PhD, is a research fellow at the University of Wollongong. His current interests include the electrophysiology of ADHD and other behavioral disorders. **Robert J. Barry**, PhD, DSc, is professor of psychology at the University of Wollongong, specializing in psychophysiology. He is a fellow of the International Organization of Psychophysiology, a member of its board of directors, and chair of its Ethics Committee. He was founding president of the Australasian Society for Psychophysiology. **Rory McCarthy**, MB, ChB, FRACP, is a consultant developmental pediatrician specializing in learning disabilities, ADHD, pervasive developmental disorders, and behavioral disorders. **Mark Selikowitz**, MB, FRACP, is a consultant developmental pediatrician in private practice. His current interests include learning disabilities, ADHD, behavioral disorders of childhood, and Asperger syndrome. Address: Adam R. Clarke, Department of Psychology, University of Wollongong, NSW, 2522, Australia; e-mail: adam_clarke@now.edu.au

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